

Financial Data Sharing Based on Cloud Computing Security: the Application and Effectiveness of Homomorphic Encryption Technology

Ruiqi Rao

Shanghai University of Finance and Economics

Shanghai, China

202311118@stu.sufe.edu.cn

Abstract—With the deep penetration of cloud computing in the financial industry, the convergence amount of financial data in the cloud continues to climb, and the problem of data security has become a bottleneck restricting the development of the financial data sharing. Traditional security protection means, such as access control, data encryption and storage, etc., gradually expose their limitations in the face of increasingly complex network attacks and diversified data use scenarios, and homomorphic encryption technology arises at the historic moment. This paper constructs a financial data security sharing scheme including financial institutions, financial certification center, Credit Data Statistics Center and financial key management center. The scheme first encrypts customer credit data and uploads it, then computes the encrypted data, and finally returns the ciphertext result to be decrypted by financial institutions. It not only ensures data privacy, but also realizes data sharing, and provides practical methods for financial institutions to prevent non-performing loans. As standardization efforts progress and computational overheads diminish, homomorphic encryption is expected to underpin next-generation secure financial infrastructures, fostering innovation in personalized financial services and regulatory compliance frameworks.

Index Terms—Financial data sharing; Cloud computing security, Homomorphic encryption technology, Data security risks, Optimization of encryption algorithms.

I. INTRODUCTION

A. Status of financial data sharing

With the rapid development of cloud computing technology [1], the financial industry actively makes use of the powerful ability of cloud computing to realize the high efficiency and convenience of data storage [2], processing and analysis [3]. However, with the massive convergence of financial data in the cloud [4], [5], security issues have become increasingly prominent [6], [7], which has become a key factor restricting the in-depth application of cloud computing in the financial field [8], [9]. Financial data includes customers' personal identity information, financial status, transaction records and other sensitive contents. Once leaked, it will not only cause huge economic losses and privacy infringement to customers [10], but it may also cause a crisis of trust in financial institutions and have an impact on the stability of the entire financial system [11]. Traditional cloud computing security protection means, such as access control, data encryption and storage, etc., gradually expose their limitations in the face

of increasingly complex network attacks and diversified data use scenarios. For example, in the process of data calculation and processing, when encrypted data is needed to operate, the traditional encryption method often needs to decrypt first, which makes the data at risk of being stolen or tampered with after decryption.

B. Characteristics and application value of homomorphic encryption technology

Homomorphic encryption, a novel cryptographic technology, offers innovative solutions for financial data security in cloud computing. It allows direct calculations on encrypted data, with the decrypted results being the same as those from plaintext computations [12]. This keeps financial data encrypted during cloud-based processing, minimizing the risk of data leakage. In the context of financial data sharing, traditional models face hurdles. Financial institutions worry about core data leakage and loss of competitiveness when sharing data, and sharing plain data directly exposes it to malicious attacks during transmission and processing [13] [14]. With homomorphic encryption, institutions encrypt customer data before uploading it to the cloud. In the cloud, encrypted data can be processed, like calculating loan totals across institutions or the sum of overdue times. Afterward, the encrypted results are sent back for decryption using private keys. The cloud platform cannot access the real data, ensuring data privacy. The use of homomorphic encryption in cloud computing security not only addresses financial data sharing privacy concerns, but also enables deeper cooperation among financial institutions [15]. By jointly analyzing and computing encrypted data, they can refine risk assessment models, optimize credit decision-making, and boost overall risk management [16]. For cloud computing service providers, adopting this technology enhances service security and appeal, meeting the financial industry's stringent data-security and privacy demands and helping expand their market share.

II. PROBLEMS WITH FINANCIAL DATA SHARING

In the context of the wide application of cloud computing technology, although financial data sharing has developed with the help of it, a series of problems have emerged, which have a

serious impact on the security, accuracy and sharing efficiency of financial data, which are shown as follows:

A. The data trading sharing mode

In the data transaction sharing mode, buyer institutions encounter several problems. High access and development costs are a major issue. As different institutions have diverse data interface specifications and technical standards, buyer institutions must conduct targeted access development each time they seek data, adding significant development and time costs and impeding efficient financial data sharing. The data pricing mechanism is unreasonable. Sellers control the pricing power, leaving buyers with no say and their rights unprotected. This leads to prices deviating from the reasonable range, dampening buyer enthusiasm and harming the healthy and fair development of the financial data sharing market. There is also a risk of data abuse. Buyer-obtained original customer credit data may be resold, violating customer privacy, disrupting the financial market, threatening data security and controllability, and worsening the trust crisis among financial institutions. Buyer institutions may also maliciously leak or seize customer resources using competitors' core data, breaching market ethics and undermining the data sharing environment. Moreover, the current data trading sharing mode lacks a proper quality testing and supervision system [17]. This makes it hard to ensure data quality, and low-quality data misleads financial institutions' decisions, increases credit risk assessment errors, and interferes with risk management and business development.

B. The information system sharing mode

In the mode of information system sharing, financial institutions need to upload their customer credit data to the sharing system managed by the credit investigation company. However, the degree of financial institutions' trust in the system is limited, because once the system has security vulnerabilities or suffers from malicious attacks, the institutional customer data is easy to leak [18]. This risk of data leakage makes financial institutions concerned when participating in data sharing, which greatly affects their enthusiasm to participate. Due to the fear of data leakage, financial institutions may not provide complete and real data to the sharing system, which makes the data quality difficult to guarantee. Incomplete or inaccurate data will reduce the accuracy of credit assessment, fail to provide a reliable decision-making basis for financial institutions, and seriously weaken the actual value of the sharing system. The risk of data leakage and data quality problems makes financial institutions reluctant to actively access the sharing system [19]. The limited number of institutions signed to join and the lack of integrity of the customer list are limited, which further affects the comprehensiveness and practicability of shared data and forms a vicious circle, leading to the difficulty of the sharing system to play its due role.

C. Cloud computing security

The security vulnerability of the cloud computing platform itself is a great hidden danger to the sharing of financial data.

For example, vulnerabilities and improper configuration of system software can be exploited by hackers to steal or manipulate financial data. If such a data security incident occurs, it will not only damage the interests of financial institutions and customers, but it will also have a serious impact on the reputation of cloud computing service providers. In a cloud computing environment [21], [22], multi-user data is often stored on the same physical infrastructure. Although cloud computing providers have adopted data isolation technology, the risk of incomplete data isolation cannot be completely excluded, which can lead to the mutual leakage of data from different financial institutions, which poses a serious threat to data security. Moreover, financial institutions struggle to build full trust in cloud service providers. Cloud service providers have a large amount of financial data, which could lead to data leaks [23]. In addition, the operational stability of cloud service providers will also affect the security of financial data. If providers have business problems, the data may be lost or inaccessible.

III. FINANCIAL DATA SECURITY SHARING SCHEME BASED ON HOMOMORPHIC ENCRYPTION

To solve the problem of data privacy in the sharing mode of the information system is the key to achieving financial data security sharing. Homomorphic encryption algorithm provides a feasible way for this. This paper applies the algorithm to financial data sharing, first encrypts customer credit data and uploads it, then computes the encrypted data, and finally returns the ciphertext result to be decrypted by financial institutions. This not only ensures data privacy, but also realizes data sharing, and provides practical methods for financial institutions to prevent non-performing loans. Li et al. [22] proposed a novel contrastive learning method for money laundering detection, which combined machine learning approaches [24]–[26] to defend data systems. When designing the financial data security sharing scheme, it is necessary to comprehensively consider the general demands of multi-party data sharing and the special requirements of financial data sharing for privacy protection, conduct in-depth analysis on the characteristic requirements of financial data security sharing, and then establish the following design objectives.

1) *Data validity*: A strict identity authentication mechanism is essential to ensure only legitimate financial institutions can share data, guaranteeing data authenticity from the start.

2) *Data confidentiality*: Protecting the privacy of financial institution customers' data during transmission, storage, and analysis is a key goal. Advanced encryption and management mechanisms safeguard data confidentiality and prevent leaks.

3) *Correctness of results*: Precise multi - party customer data statistics are a basic requirement. The scheme should use proper algorithms and logic, and undergo strict testing, to ensure accurate results for financial decision - making.

4) *Platform independence*: The solution should be widely applicable, not tied to specific platforms or technologies. Any platform with a suitable homomorphic encryption environment

and computing power can operate, enhancing its versatility and scalability.

5) *Efficiency*: To ensure financial data sharing is efficient and practical, the scheme should use high - efficiency homomorphic encryption algorithms. Optimizing the algorithm reduces processing time and improves sharing efficiency while maintaining data security.

A. Structural design

In order to use homomorphic encryption for data sharing, it is necessary to set up an institution responsible for the summary calculation of ciphertext data, which is called the Credit Data Statistics Center. At the same time, to ensure the privacy of the data, the institution is not allowed to decrypt the data and may not be able to grasp the homomorphic key information. It is necessary to set up another institution to manage the homomorphic key; this institution is called the Financial Key Management Center. In order to ensure the validity of the data, it is necessary to set up an authoritative certification body responsible for managing the legal identity of all participating financial institutions and Credit Data Statistics Centers, which is called the Financial Certification Center. To sum up, the plan includes four participating entities: financial institutions, Financial Certification Center, Credit Data Statistics Center and Financial Key Management Center. The authentication center is used to generate the identification code and digital signature key of the legitimate financial institution; the Financial Key Management Center is used to generate the same batch for the participating financial institution; the financial institution encrypts its customer credit data locally and submits it to the Credit Data Statistics Center; the Credit Data Statistics Center calculates the ciphertext data submitted by all institutions and finally returns the ciphertext results to each institution to decrypt the shared customer data results. The overall framework of the scheme is shown in Figure 1. In this scheme, the Financial Certification Center is only responsible for the generation of identity certificate and digital signature keys, and will not access the customer data of financial institutions; the Financial Key Management Center is only responsible for the generation of key pairs and cannot obtain the data to be encrypted, and the Credit Data Statistics Center cannot decrypt the data. Therefore, there is no possibility of data leakage in the process of transmission, calculation and storage, which ensures the security of the data sharing process. At the same time, financial institutions and Credit Data Statistics Center need to digital sign and verify the data to ensure the validity and immutability of the data, and ensure the quality of the shared data ultimately obtained by financial institutions. The main work responsible for by the individual participating entities in the programme is shown in Table 1. In this scheme, the Financial Certification Center is only responsible for the generation of identity certificate and digital signature keys, and will not access the customer data of financial institutions; the Financial Key Management Center is only responsible for the generation of key pairs and cannot obtain the data to be encrypted, and the Credit Data

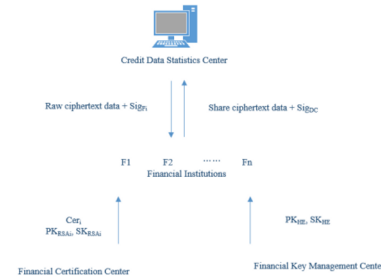


Fig. 1. framework of the scheme

Entity	Main work
Financial Institutions, F	Registration/login authentication, Initiate/accept team invitations, Request homomorphic public and private keys, Input data; Encryption of customer's original credit data, Digital signature/visa verification; Decrypt shared data ciphertext
Financial Certification Center, CC	Generate ID codes for legitimate financial institutions, Generate figures for financial institutions and credit data centers, Signature key
Financial Key Management Center, KC	Set key parameters, Generate homomorphic cryptographic public and private keys, Enable/disable key
Credit Data Statistics Center, DC	Digital signature/visa verification, Receive and calculate ciphertext data from legitimate financial institutions; Submit calculated ciphertext data to financial institutions

TABLE 1

PARTICIPATING ENTITIES AND MAJOR ACTIVITIES

Statistics Center cannot decrypt the data. Therefore, there is no possibility of data leakage in the process of transmission, calculation and storage, which ensures the security of the data sharing process. At the same time, financial institutions and Credit Data Statistics Center need to digital sign and verify the data to ensure the validity and immutability of the data, and ensure the quality of the shared data ultimately obtained by financial institutions. The main work responsible for by the individual participating entities in the program is shown in Table 1.

B. Process design

The process of the financial data security sharing scheme based on homomorphic encryption is divided into five stages: initialization, team formation and key application, encryption and transmission, ciphertext calculation, and decryption sharing. The initialization stage is mainly to generate the legal financial institution identity certificate, set the digital signature key and arrange the homomorphic encryption environment. At the stage of team formation and key application, legal financial

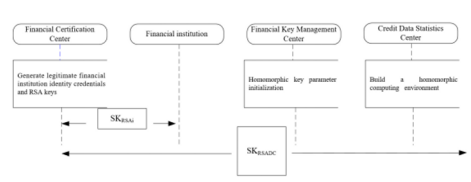


Fig. 2. Initialization phase diagram

institutions are accepted for system registration, data sharing team formation and homomorphic encryption key distribution; In the encryption and transmission phase, each institution encrypts its own customer credit data and submits it to the Credit Data Statistics Center; In the ciphertext calculation stage, the Credit Data Statistics Center will conduct ciphertext calculation after checking the data sent by various institutions; In the decryption sharing stage, financial institutions check and decrypt the calculated ciphertext data to obtain the customer data sharing result. The stages are described in detail below.

1) *Initialization stage*: Financial Certification Center for every legal financial institution $F_i (i \in 1, 2, 3, \dots, n)$ Assign a unique and unalterable registration certificate code (this simulated identity certificate is a 32-bit length string, which randomly consists of "a-z", "0-9") to prepare for the legitimacy of the subsequent verification institution. At the same time, the Financial Certification Center is to generate the unique corresponding digital signature public key PK_{RSAi} and private key SK_{RSAi} for F_i , and the Financial Certification Center generates the digital signature public key PK_{RSADC} and private key SK_{RSADC} for the Credit Data Statistics Center (DC). The private key is distributed to the financial institution and the DC for signing the ciphertext data, and the public key is used by the financial institution and the DC for verifying the signature of the received ciphertext data. In addition, the Financial Key Management Center and the Credit Data Statistics Center will initialize the homomorphic key parameters and build the homomorphic computing environment Env_{HE} .

The stage flow is shown in Figure 2.

2) *Team formation and key application stage*:

- **Team formation**:
 F_i , the financial institution participating in the sharing, enters the data sharing system for registration with its identity certificate Cer_i . After receiving the request, the system authenticates the identity of F_i through comparison, ensuring that the registered Cer that only meets the conditions can log in the system. Financial institutions that successfully log in to the system select other institutions that need to form a team and invite them. After the invited organization sees the team request information, it can choose whether to accept the invitation according to the needs.
- **Key application**:
After the completion of the team, the institution initiates a key application to the Financial Key Management Center; the Financial Key Management Center generates homo-

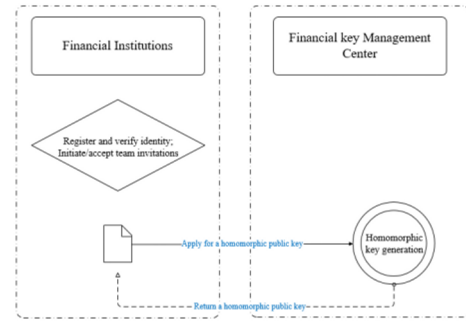


Fig. 3. Team formation and key application diagram

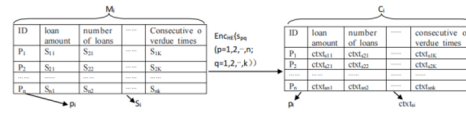


Fig. 4. Customer data encryption process for financial institution F_i

morphic encryption public key Management Center, the financial key management center generates homomorphic encryption public keys PK_{HE} and private keys SK_{HE} for the team, and returns the PK_{HE} to all participating institutions of the team for the next stage of encrypted data use, and the SK_{HE} is returned to the institution in the decryption stage for security reasons. The stage flow is shown in Figure 3.

3) *Encryption and transmission stage*:

- **Encryption**:
The financial institution F_i performs homomorphic encryption operations on the customer data $M_i = \{P_i, S_i\}$ to be shared. The first field of the data M_i is the customer ID (each ID number corresponds to an ID, and the ID of the same customer in different machines is the same), followed by k word segments for the customer's credit data.
$$S_i = \begin{pmatrix} S_{11} & \cdots & S_{1k} \\ \vdots & \vdots & \vdots \\ S_{n1} & \cdots & S_{nk} \end{pmatrix} (S_{pq} \in R, p = 1, 2, \dots, n, q = 1, 2, \dots, k)$$
Keep the ID unchanged, encrypt each credit data S_{pq} with homomorphic public keys PK_{HE} successively to obtain ciphertext data $ctxt_{spq} = Enc(S_{pq}, PK_{HE})$ (Enc_{HE} is the encryption algorithm), then the customer data changes from M_i to $C_i = \{P_i, Ctxt_i\}$. Figure 4 is a schematic diagram of the customer data encryption process of a financial institution participating in data sharing.
- **Transmission**:
 $MD5$ is used to generate the information digest $InfoSum_{C_i} = H_{MD5}(C_i)$ of C_i , and SK_{RSAi} is used to digitally sign the information digest to obtain the signature information $Sig_i = Sig(InfoSum_{C_i}, SK_{RSAi})$ ($MD5$ algorithm is used to generate the hash value in the digital signature, Sig is the signature algorithm), and

then ciphertext data, information digest and signature information $\{C_i, InfoSum_{C_i}, Sig_i\}$ are submitted to the credit Data Statistics Center.

4) *The ciphertext calculation stage:* After receiving the information from various institutions, the credit data statistics Center first verifies the signature information with PK_{RSAi} to confirm that the information belongs to a legitimate financial institution and has not been tampered with. After the verification is passed, the k field data of each data record with the same ID in all ciphertext is summed and calculated to obtain ciphertext data $Ctxt_{HE} = \{Ctxt_{ALL}, P_{ALL}\}$ composed of all customer data, that is, the credit data of the same customer of all financial institutions is integrated, and then the ciphertext information, information summary and signature information $\{Ctxt_{HE}, InfoSum_{Ctxt_{HE}}, Sig_{DC}\}$ are sent to each financial institution. The process of digital signature sent to various financial institutions is similar to that used in 3), so it will not be repeated here.

5) *Decryption and sharing stage:* Financial institution F_i uses 1 to check the $Ctxt_{HE}$ sent by the credit data Statistics Center, and sends a request to the financial key management center after the verification is passed. The financial key management center will distribute the homomorphic private key to all participating institutions in the team, and financial institution F_i uses SK_{HE} to decrypt $Ctxt_{HE}$ and get the plaintext result M_{ALL} .

Through the above steps, the multi-party data can be analyzed, summarized and calculated under the encrypted conditions, and then the calculated ciphertext is returned to each organization. Finally, the organization can view the integrated and calculated customer data through the homomorphic key.

IV. EFFICIENCY STUDY OF HOMOMORPHIC ENCRYPTION ALGORITHMS FOR FINANCIAL DATA

A. Experimental environment

In this section, the efficiency of Paillier homomorphic encryption algorithm is studied, and the relationship between the running time of each stage and the number of key bits, the amount of data and the number of data bits is analyzed through experiments. The experimental environment in this paper is 11th Gen Intel(R) Core(TM) i5-1155G7 @ 2.50GHz 2.50 GHz. The IDE is Visual Studio Code 1.82.2, the program programming languages are Python and C++. In addition, due to the system scheduling and CPU dynamic frequency modulation problems, the running time of the program will be different. The experimental results are the average values of 50 experiments after removing the maximum and minimum values.

B. Relationship between algorithm efficiency and number of key bits

The number of key bits determines the length of public key n (in binary bits), which directly affects the performance of encryption, decryption and ciphertext computation. The experiment kept the amount of encrypted data (100 32-bit random integers each time) and the number of data bits

Number of bits	encryption	addition homomorphic algorithm	decryption
8	703	2	1241
16	711	2	1264
32	717	2	1259
64	742	2	1246
128	786	2	1287
256	847	2	1268
512	1013	2	1237
1024	1296	2	1276

TABLE II
PAILLIER ALGORITHM RUNNING TIME AS A FUNCTION OF THE NUMBER OF DIGITS

constant, and measured the running time of key generation, encryption, addition homomorphic calculation and decryption under different key bits. The results are shown in Table 2. The experimental results show that:

- Key Generation : As the number of key bits increases, the time required for key generation increases significantly. Especially at 2048-bit and above, the time overhead of key generation is large.
- Encryption process : The encryption time increases non-linearly with the number of key bits, mainly due to the high computational complexity of the encryption process, which involves large integer powers.
- Homomorphic computation of addition : The time cost of the homomorphic operation of addition is relatively small, and the impact of the number of key bits is not as significant as encryption and decryption.
- Decryption process : Decryption time increases the most, especially above 1024-bit. This is because the decryption process involves modular operation and large number calculation, and the amount of calculation increases sharply with the growth of the key length.

In summary, in order to strike a balance between security and efficiency, it is recommended to use 1024-bit or 2048-bit keys in financial data sharing scenarios to ensure appropriate computational performance.

C. Relationship between algorithm efficiency and data volume

Encrypting or decrypting n data requires n independent computations, so its time complexity is $O(n)$, which is linear in the amount of data. In the experiment, the number of key bits was 1024-bit, the number of data bits was 32-bit, and the running time was tested under different data amounts (100 to 10,000). The results are shown in Fig 5, The experimental results show that:

- Encryption time : increases linearly with the amount of data. When the data size reaches 10,000, the encryption takes about 70 seconds, showing the scalability of Paillier algorithm in large data volume scenarios.
- Decryption time : The growth trend is similar to encryption, with decryption taking about 1.5 times as long as encryption.

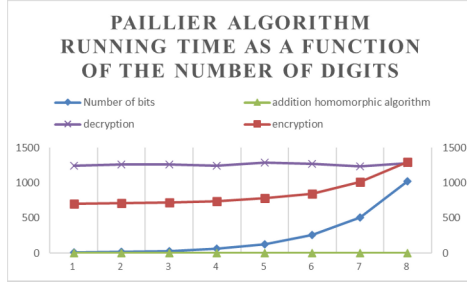


Fig. 5. Paillier algorithm running time as a function of the number of digits

- Homomorphic addition : Homomorphic addition is less time-consuming than encryption and decryption, and can be done efficiently even for large data sets.

These results show that Paillier algorithm can still maintain reasonable performance in large-scale data processing scenarios and is suitable for batch computing requirements of financial data sharing.

V. CONCLUSIONS AND FUTURE WORKS

Homomorphic encryption technology provides an innovative solution to secure financial data exchange, and its potential is far from fully realized. With the deepening of technology iteration and application scenarios, further exploration and expansion can be carried out in the following directions in the future to promote financial data sharing to a higher level of intelligence, collaboration, and globalization.

1) Breakthrough improvement of algorithm performance: Combined with emerging technologies such as quantum computing and edge computing, the parallel computing capability of the homomorphic encryption algorithm is optimized to achieve a millisecond-level encryption operation response. Develop a customized algorithm library for financial scenarios, support real-time crypto aggregation and cross-institutional joint analysis of large-scale data sets, and help the comprehensive crypto transformation of high-frequency trading, real-time risk control and other business scenarios.

2) Cross-modal data fusion and value mining: Break through the limitation of single structured data, explore the homomorphic encryption processing technology of unstructured financial data (such as text, image, transaction behavior log). Combined with privacy-enhanced machine learning framework, feature extraction and joint modeling of multi-modal data are realized in the encrypted state, providing richer decision support for precision marketing, anti-money laundering and other scenarios.

3) Green Encryption and Sustainable Computing Develop low-energy homomorphic encryption algorithms, optimize hardware acceleration schemes (such as TPU/ASIC chips), and reduce the dependence of encryption computing on energy and computing power. Explore the *Crypto-as-a-Service* (EaaS) business model and enable flexible resource allocation through cloud-native architecture to help financial institutions achieve carbon neutrality while ensuring data security.

REFERENCES

- [1] WEN Jie, ZHANG Zhixia, LAN Yang, et al., A survey on federated learning: challenges and applications[J]. International Journal of Machine Learning and Cybernetics, 2023, pp.513–535.
- [2] Jin X, Zhang H, et al., Confused-modulo-projection-based somewhat homomorphic encryption-Cryptosystem, library, and applications on secure smart cities. IEEE IoT Journal, 2020, pp.6324–6336
- [3] Li X, Gao H, Zhang J, et al., GPU accelerated full homomorphic encryption library and cryptosystem, applications for IoT systems. IEEE Internet of Things Journal, 2023, pp.6893–6903
- [4] M. Qiu and H. Qiu, Review on image processing based adversarial example defenses in computer vision, IEEE 6th BigDataSecurity, 2020
- [5] H. Lu, X. Wang, et al., The effects of using chaotic map on improving the performance of multiobjective evolutionary algorithms, Math. Problems in Eng. 2014 (1), 924652, 2014
- [6] K. Gai, et al., Blockchain-enabled service optimizations in supply chain digital twin, IEEE T. Serv. Comp. 16 (3), 1673–1685, 2022
- [7] Y. Qiu, et al., A Closer Look at GAN Priors: Exploiting Intermediate Features for Enhanced Model Inversion Attacks. ECCV (32) 2024, pp. 109–126
- [8] Li Yashuo, Long Chun, et al., Privacy protection method for face recognition based on Homomorphic encryption. Infor. Security Res., 2023, pp.843–850
- [9] LI Zhang, XU Jianbo, VIJAYAKUMAR P, et al., Homomorphic encryption-based privacy-preserving federated learning in IoT-enabled healthcare system, IEEE T. Network Sci. and Eng., 2022, pp. 2864–2880.
- [10] Bonawitz K, Ivanov V, et al. Practical secure aggregation for privacy-preserving machine learning, ACM CCS 2017: pp.1175–1191
- [11] TAMILARASI G, GANDHI K R, and PALANISAMY V. Improved Homomorphic Encryption with Optimal Key Generation Technique for VANETs. Intell. Automation & Soft Computing, 2022, 33(2).
- [12] WIBAWA F, CATAK F O, KUZLU M, et al. Homomorphic encryption and federated learning based privacy-preserving CNN training: Covid-19 detection usecase. European Interdisciplinary Cybersecurity Conf., Barcelona, Spain, 2022: 85–90
- [13] ZHANG Jiale, LIU Yue, WU Di, et al. VPFL: A verifiable privacy-preserving federated learning scheme for edge computing systems. Digital Comm. and Networks, 2023, pp.981–989
- [14] AN Haoyang, HE Debiao, BAO Zijian, et al. An identity-based dynamic group signature scheme for reputation evaluation systems. J. of Systems Arch., 2023, 139: 102875.
- [15] XIA Feng, LIU Haiyang, YANG Haowei, et al. BatchAggregate: Efficient Aggregation for Private Federated Learning in VANETs. IEEE T. Dependable and Secure Computing, 2024, pp.1–15.
- [16] Kanagavelu R, Li Z, Samsudin J, et al. Two phase multi-party computation enabled privacy preserving federated learning, IEEE/ACM CCGRID, 2020, pp.410–419
- [17] Aono Y, Hayashi T, Wang L, et al. Privacy-preserving deep learning via additively homomorphic encryption, IEEE T. Information Forensics and Security, 2017, pp.1333–1345
- [18] WANG Yong, Research on key technologies of secure model aggregation for federated learning, Ph. D. dissertation.
- [19] Lee E, Lee J.W., Kim Y.S., et al., Minimax approximation of sign function by composite polynomial for homomorphic comparison. IEEE T. Dependable and Secure Computing, 2022, PP.3711–3727
- [20] M. Qiu, W. Dai, A. Vasilakos, Loop parallelism maximization for multimedia data processing in mobile vehicular clouds. IEEE T. Cloud Comp., 7 (1), 250–258 2016
- [21] M. Qiu, K. Zhang, M. Huang, Usability in mobile interface browsing. Web Intelligence and Agent Systems, 4 (1), 43–59, 2006
- [22] Z. Li, J. Huang, et al., Contrastive Learning for Money Laundering Detection: Node-Subgraph-Node Method with Context Aggregation and Enhancement Strategy. KSEM (4), 2024, pp.31–47.
- [23] Cheon J. H, Kim D., Kim D., Efficient homomorphic comparison methods with optimal complexity, Cryptology ePrint Archive, Paper 2019/1234, <https://eprint.iacr.org/2019/1234>
- [24] Y. Zhang, M. Qiu, H. Gao, Communication-Efficient Stochastic Gradient Descent Ascent with Momentum Algorithms. IJCAI, 4602–4610, 2023
- [25] C. Ling, J. Jiang, et al., Deep Graph Representation Learning and Optimization for Influence Maximization, ICML 2023
- [26] Y. Zhang, B. Pan, et al., Visual Attention Prompted Prediction and Learning. IJCAI 2024: 5517–5525